

Simulator Metrics and KPI Reference Tables

The following structured tables contain the raw simulator outputs, key performance indicators (KPIs), and hardware calibration targets required for architectural evaluation and slide compilation.

Table 1: Raw Simulator Outputs

Category	Metric	Description	Unit
Counters	cycles	Total simulated clock cycles	int
	MACs	Multiply-accumulate ops executed	int
	hbm_bytes	Total bytes transferred to/from HBM	bytes
Performance	ttft_ns	Time To First Token	ns
	decode_tps	Decode throughput	tok/s
	roofline_eff	Actual perf as % of roofline bound	%
	roofline_bound	compute- or memory-limited	str
Utilization	systolic_util_pct	Avg utilization across systolic	%

Category	Metric	Description	Unit
	dma_util_pct	DMA engine utilization	%
	vec_util_pct	Vector core utilization	%
	access_util_pct	Access core utilization	%

Table 2: Key Performance Indicators (KPIs)

Category	Metric	Description / Formula	Unit
Compute	tflops_achieved	$\text{MACs} \times 2 / \text{wall_time} / 1\text{e}12$	TFLOPS
	tflops_peak	Theoretical peak from chip config	TFLOPS
	hbm_util_pct	HBM bandwidth used vs peak	%
Workload	arith_intensity	$\text{MACs} / \text{hbm_bytes}$	FLOP/byte
	mem_compute_ratio	$\text{mem_bound_cyc} / \text{compute_bound_cyc}$	ratio
Latency	ttft_per_token_ns	$\text{ttft_ns} / \text{prompt_len}$	ns/tok
	bytes_per_token	HBM bytes per token	bytes

Category	Metric	Description / Formula	Unit
Split	ffn_mac_pct	% of MACs in FFN layers	%
	attn_mac_pct	% of MACs in attention layers	%

Table 3: Hardware Calibration

Category / Hardware Target	Metric	Description / Specs	Target / Unit Value
Calibration	pub_tps_bs1	Published real-chip TPS at bs=1	tok/s
	pub_hbm_util_pct	Back-calc HBM util from pub_tps	%
	calib_hbm_err_pct	sim - pub (pass = <=20%)	delta %
H100_SXM5	hbm_bw_tb_s	3.35 TB/s 1.98 GHz TPS 120	64.5%
A100_SXM4	hbm_bw_tb_s	2.00 TB/s 1.41 GHz TPS 70	63.0%
TPUv4_chip	hbm_bw_tb_s	1.20 TB/s 0.275 GHz TPS 60	90.0%